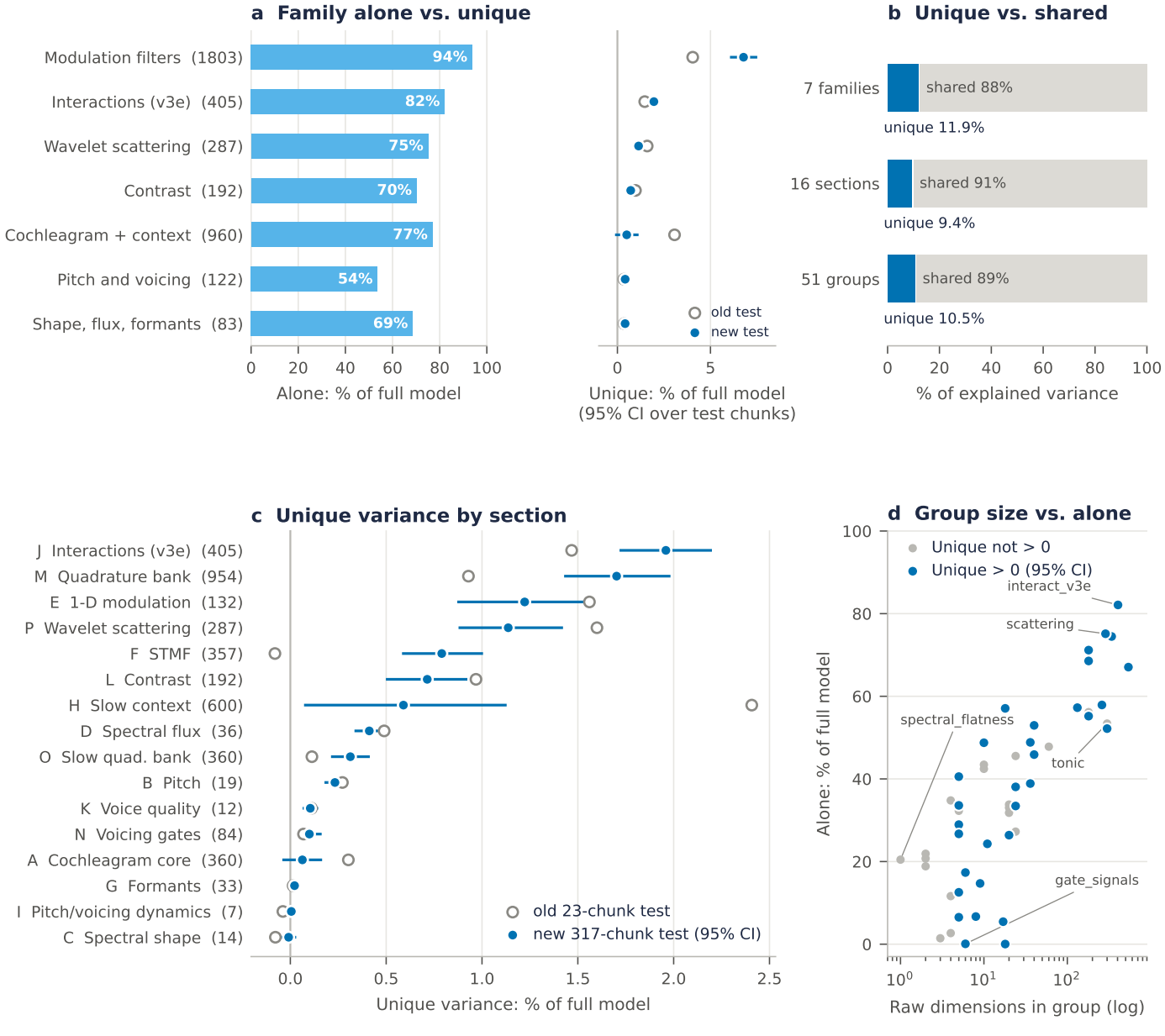
LEAN-v20 variance partitioning, new held-out test (grid; full R² 1.48%, old test 1.89%)
a Family alone vs. unique
Modulation filters (1803)
Interactions (v3e) (405)
Wavelet scattering (287)
Contrast (192)
Cochleagram + context (960)
Pitch and voicing (122)
Shape, flux, formants (83)
94%
82%
75%
70%
77%
54%
69%
0 20 40 60 80 100
Alone: % of full model
0 5
old test
new test
Unique: % of full model
(95% CI over test chunks)
b Unique vs. shared
7 families
shared 88%
unique 11.9%
16 sections
shared 91%
unique 9.4%
51 groups
shared 89%
unique 10.5%
0 20 40 60 80 100
% of explained variance
c Unique variance by section
J Interactions (v3e) (405)
M Quadrature bank (954)
E 1-D modulation (132)
P Wavelet scattering (287)
F STMF (357)
L Contrast (192)
H Slow context (600)
D Spectral flux (36)
O Slow quad. bank (360)
B Pitch (19)
K Voice quality (12)
N Voicing gates (84)
A Cochleagram core (360)
G Formants (33)
I Pitch/voicing dynamics (7)
C Spectral shape (14)
old 23-chunk test
new 317-chunk test (95% CI)
0.0 0.5 1.0 1.5 2.0 2.5
Unique variance: % of full model
d Group size vs. alone
Unique not > 0
Unique > 0 (95% CI)
interact_v3e
scattering
spectral_flatness
tonic
gate_signals
Alone: % of full model
0 20 40 60 80 100
10⁰ 10¹ 10²
Raw dimensions in group (log)